Arduino And Javascript

Click to ⭐ the Project

WhoAmI

voidash

github.com/sudhansu

Arduino

- A company in Italy
- Arduino is referred to physical programmable circuit board (microcontoller)
- ArduinoIDE

Agenda

Without Arduino Ecosystem

- Tooling is hard as it requires special programming skills and deep knowledge on how chip works and how to communicate with it.
- How to do without Arduino IDE
  - Read the manual
  - While programming disable std features and define the target architecture or download one
  - or program in assembly and use assembler
  - Then use serial interface such as RS-232 to flash the chip.

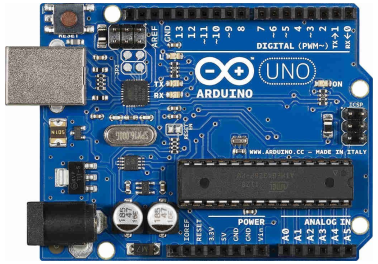
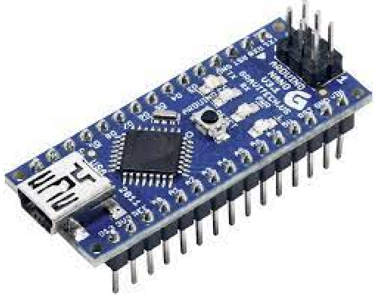
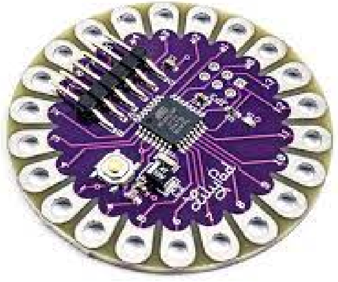
Why Arduino

- Great for prototyping
- cross platform
- Has a very low power requirements
- Is open source with large community
- Doesn't require a degree worth of studying to program it.
- Very Cheap

What is possible with Arduino

- Insert Some projects here

Arduino Flavors

				
Arduino Uno	Arduino Mega	Arduino Nano	Arduino Leonardo	Arduino lilypad

Getting the Hardware

- If you are in Nepal
  https://himalayansolution.com/product/arduino-starter-kit-1
- Access to International Payment then
  https://www.aliexpress.com/item/1005001653349193.html

Stores in Nepal

- Supreme Light Technology: https://www.sltech.com.np/
- Himalayan Solution https://himalayansolution.com/

Microcontroller and Microprocessor

- Microcontroller is a chip that houses CPU, Memory(RAM ROM), I/O controllers all in a single package however Microprocessor however only has CPU
- A computer basic operation is `input -> process -> output`
- Atmega328P has 32kb Of program Memory, 1kb of ROM and 2Kb of RAM

<table>
<tr><th>Input</th><th>Process</th><th>Output</th></tr>
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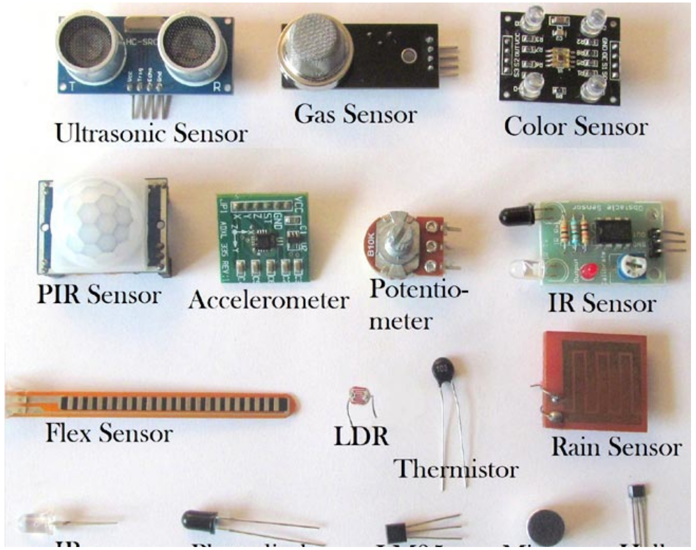
<td>
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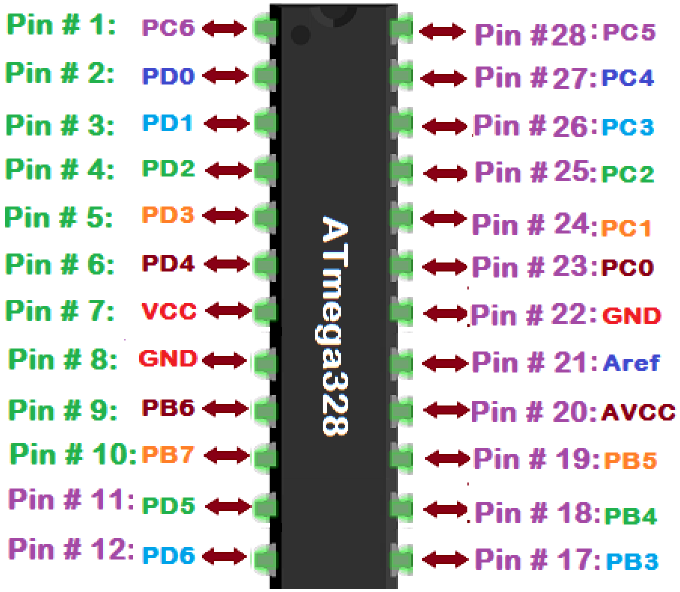
<td>
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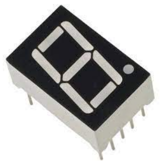
<td>
Servo Motor

7 Segment Display

Light Emitting Diodes

Relay

Speaker

LCD Display</td>
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Arduino Board

<table>
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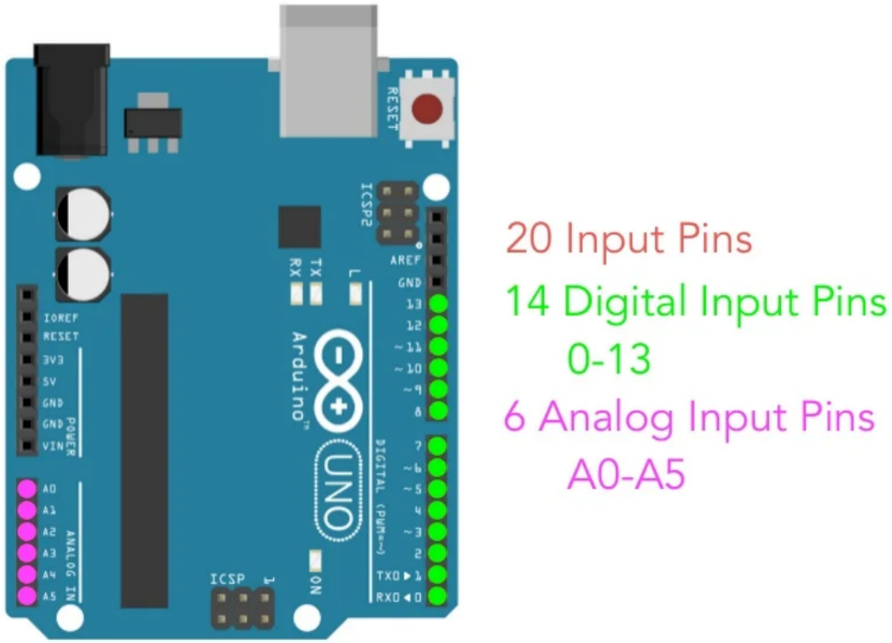

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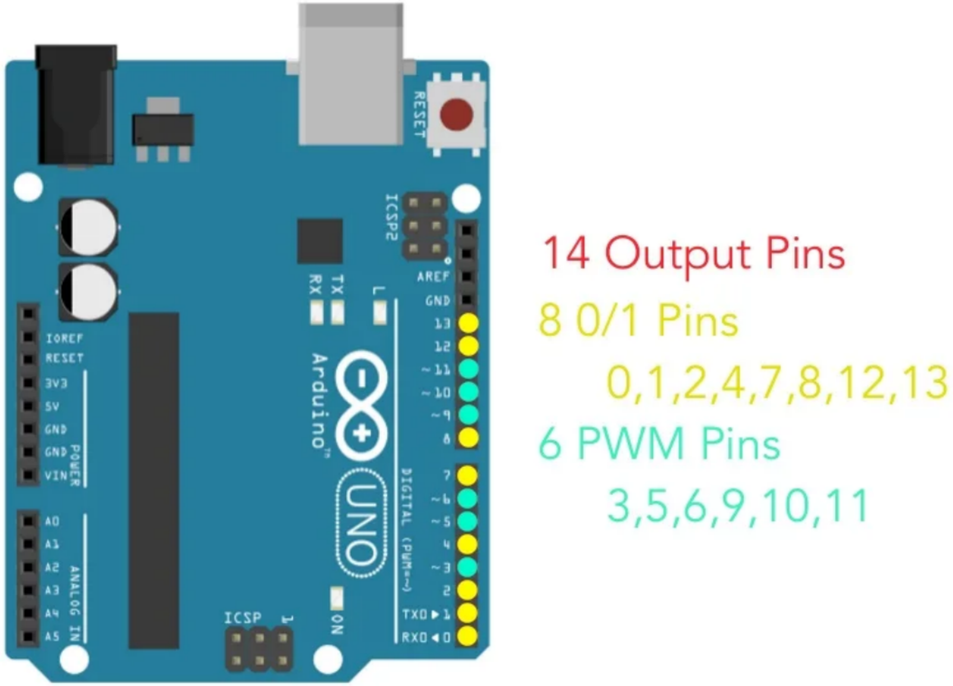


</td>
</tr>
</table>

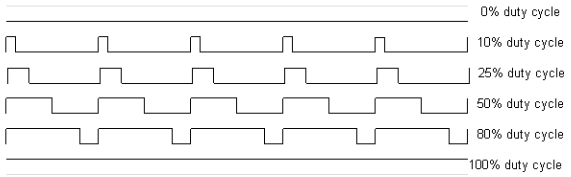
Arduino Board Continued

For Input

- Analog Pins : Used to read analog sensor data. i.e Potentiometer, Pressure Sensor, Accelerometer, LDR sensor
- Digital Pins : If sensor already has A/D converter or the output of the sensor is discrete then Digital pins are used. i.e Ultrasonic Sensor, Ball Switch Sensor. It can only detect OV or (3.3V/5V).

For Output

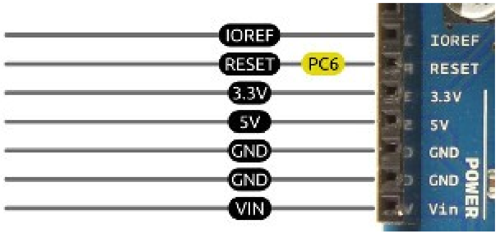
- PWM Pins : Pulse Width Modulation can be thought as a square wave continiously being on and off, where on time is the pulse width and it defines duty cycle.



- Digital Pins: They either provide 0V as off or 5V as On.

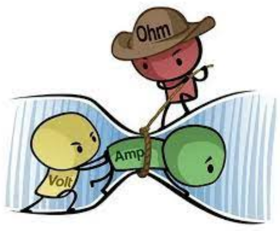
Arduino Board continued

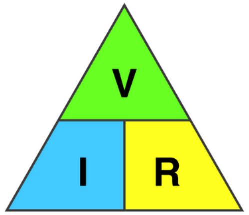
Power Pins

test	ok
	• IOREF: Ioreference • RESET: Provide 5V to reset the board • 3.3V: 3.3V output for sensors • 5V: 5V output for sensors • GND: Ground Pin • VIN: Provide 5V to power the Arduino Uno Board

Basics of Electronics

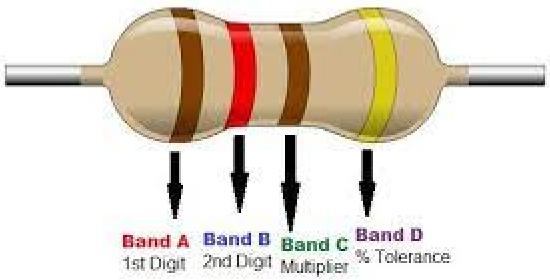
- Ohm's law $V = IR$





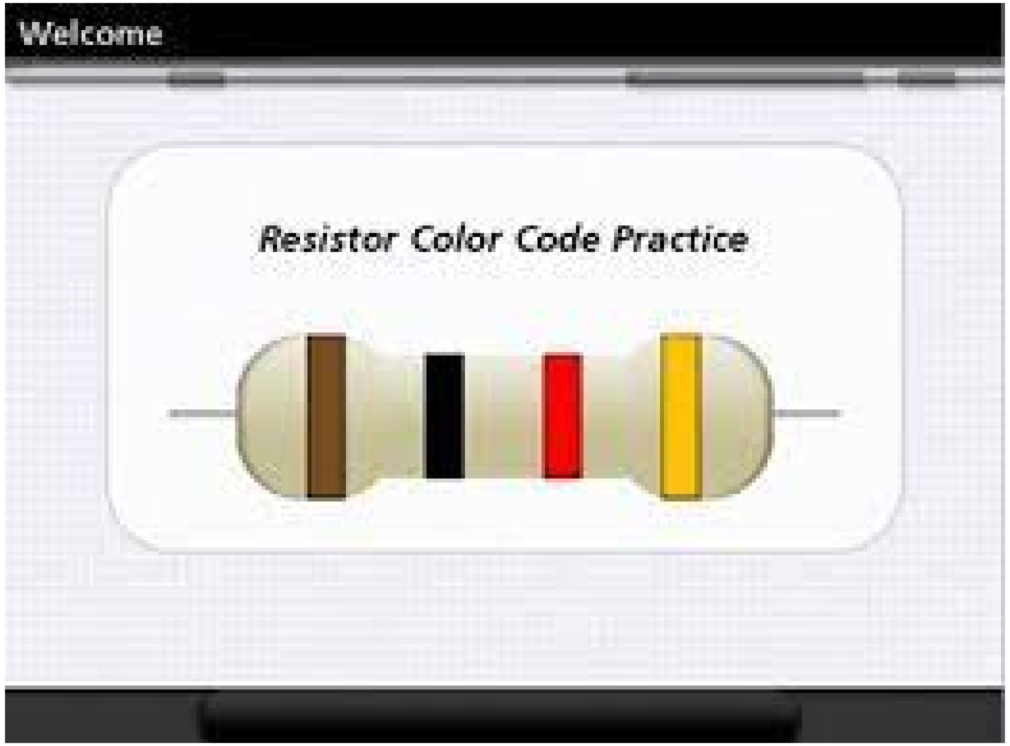
- Voltage : Electric Pressure
- Current : Flow of electrons
- Resistance : Measure of opposition of current flow

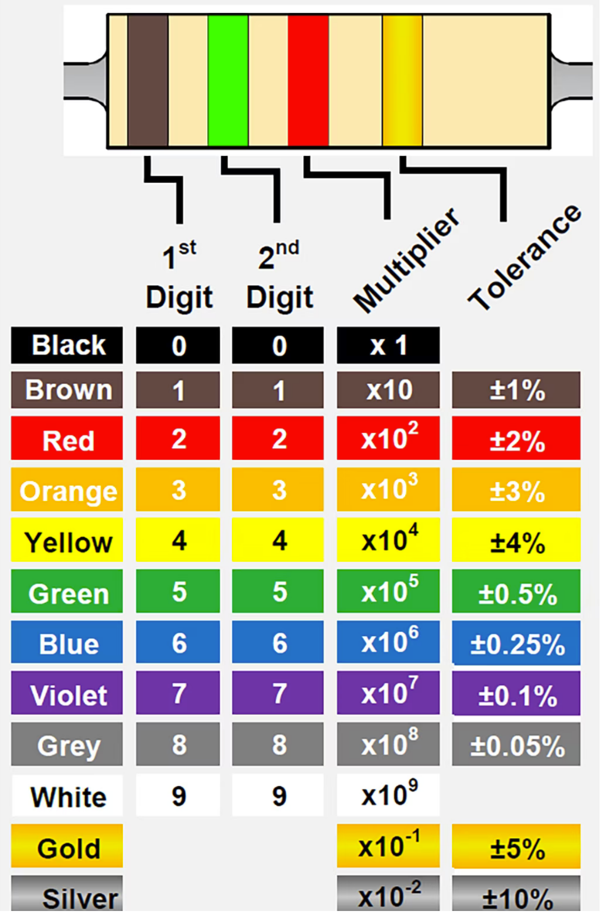
Resistor



Color	Numeric Value	Multiplier	Tolerance	Temperature coefficient
BLACK	0	1Ω		250
BROWN	1	10 Ω	±1%	100
RED	2	100 Ω	±2%	50
ORANGE	3	1K Ω		15
YELLOW	4	10 Ω		25
GREEN	5	100 Ω	±0.5%	20
BLUE	6	1M Ω	±0.25%	10
VIOLET	7		±0.1%	5
GREY	8			1
WHITE	9			
GOLD			±5%	
SILVER			±10%	

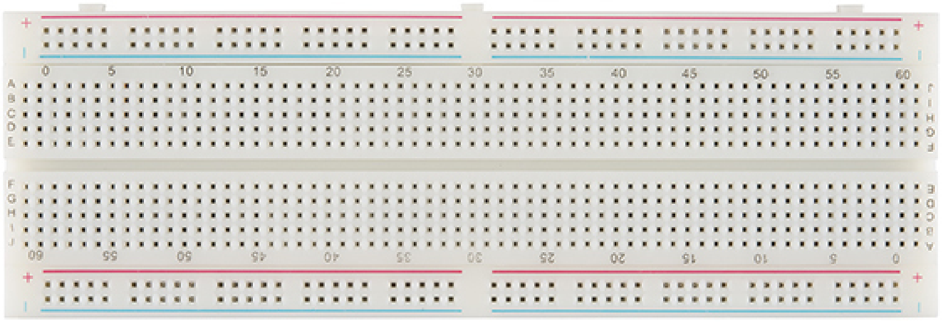
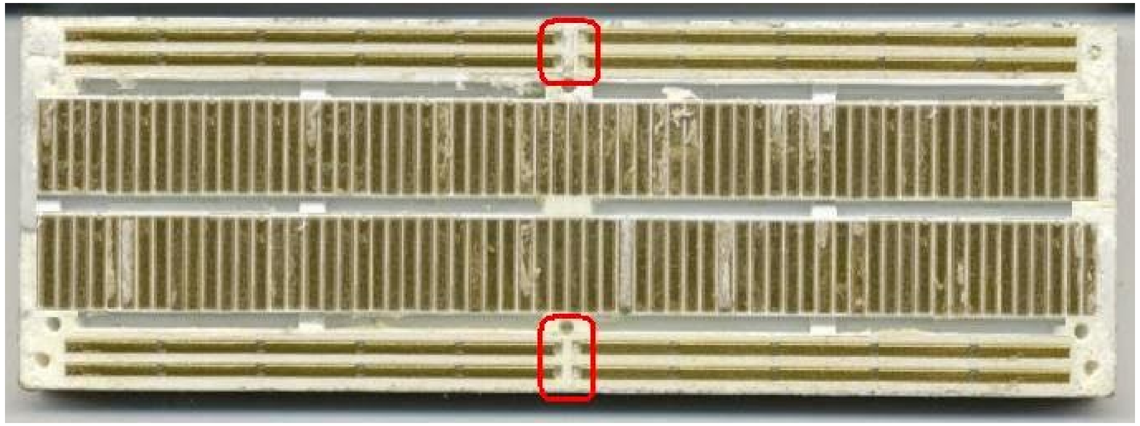
Exercise


Welcome
Resistor Color Code Practice

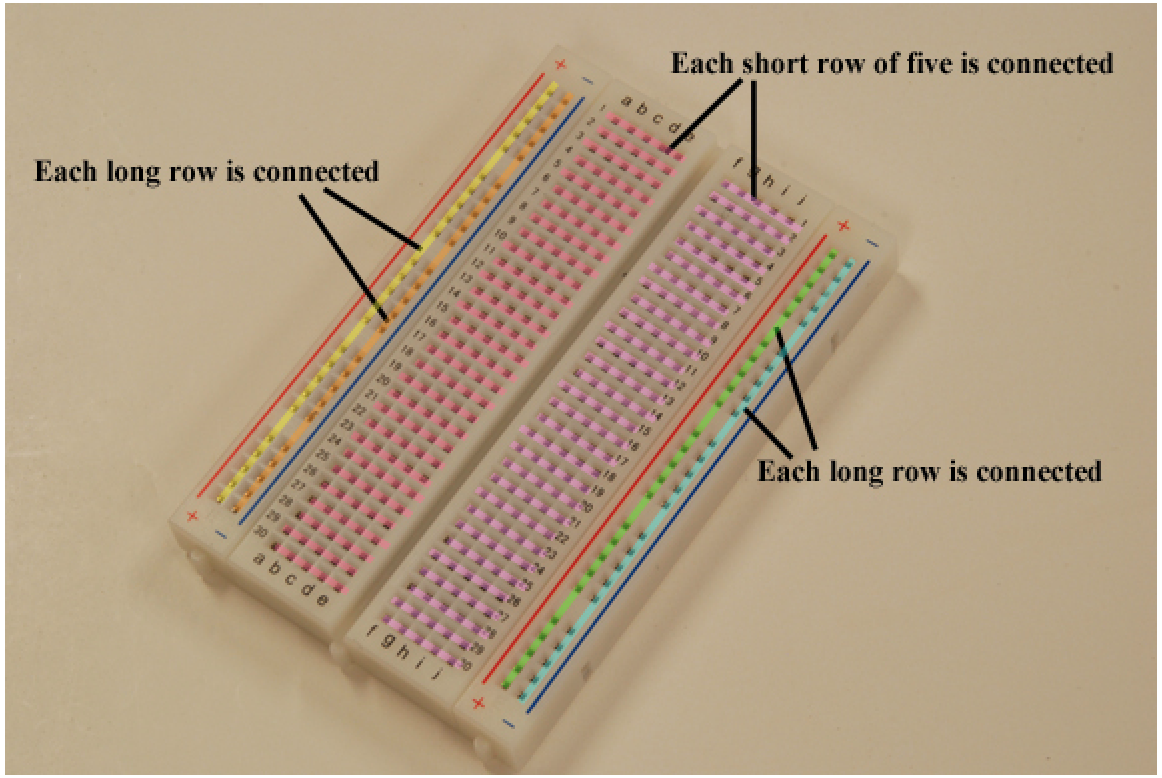

1st Digit
2nd Digit
Multiplier
Tolerance
Black 0 0 x 1
Brown 1 1 x10 ±1%
Red 2 2 x10^2 ±2%
Orange 3 3 x10^3 ±3%
Yellow 4 4 x10^4 ±4%
Green 5 5 x10^5 ±0.5%
Blue 6 6 x10^6 ±0.25%
Violet 7 7 x10^7 ±0.1%
Grey 8 8 x10^8 ±0.05%
White 9 9 x10^9
Gold x10^-1 ±5%
Silver x10^-2 ±10%

Breadboard

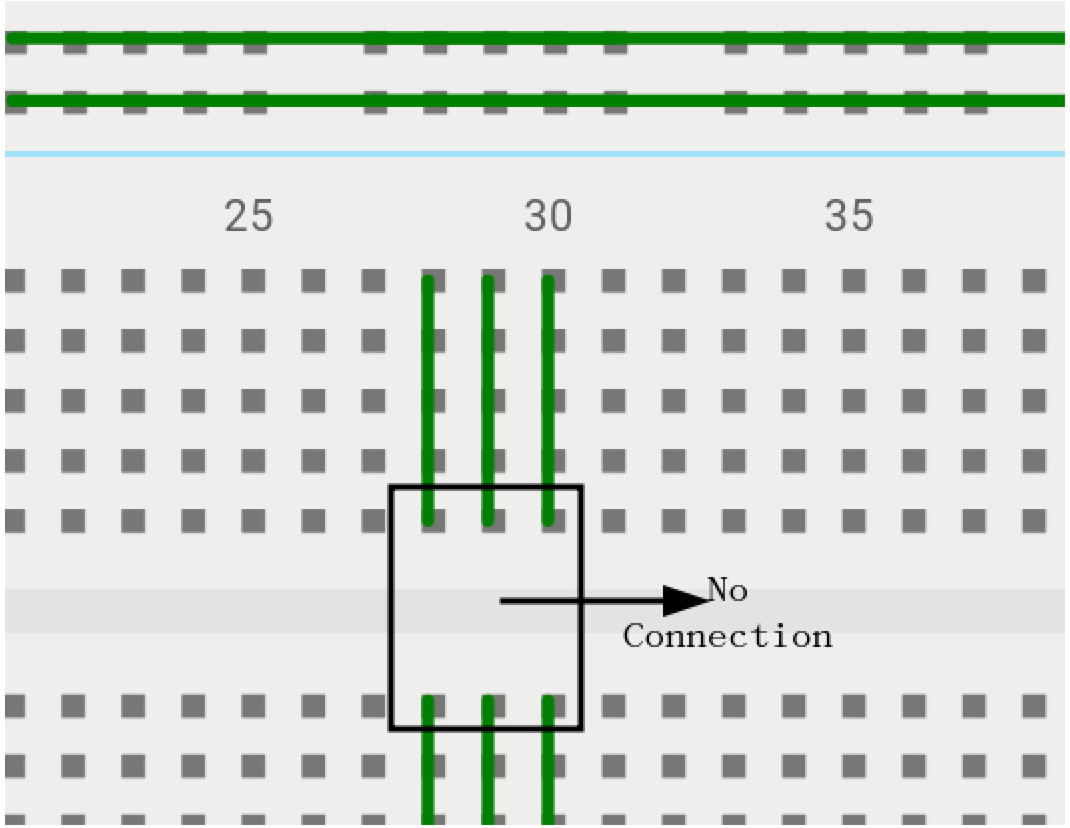
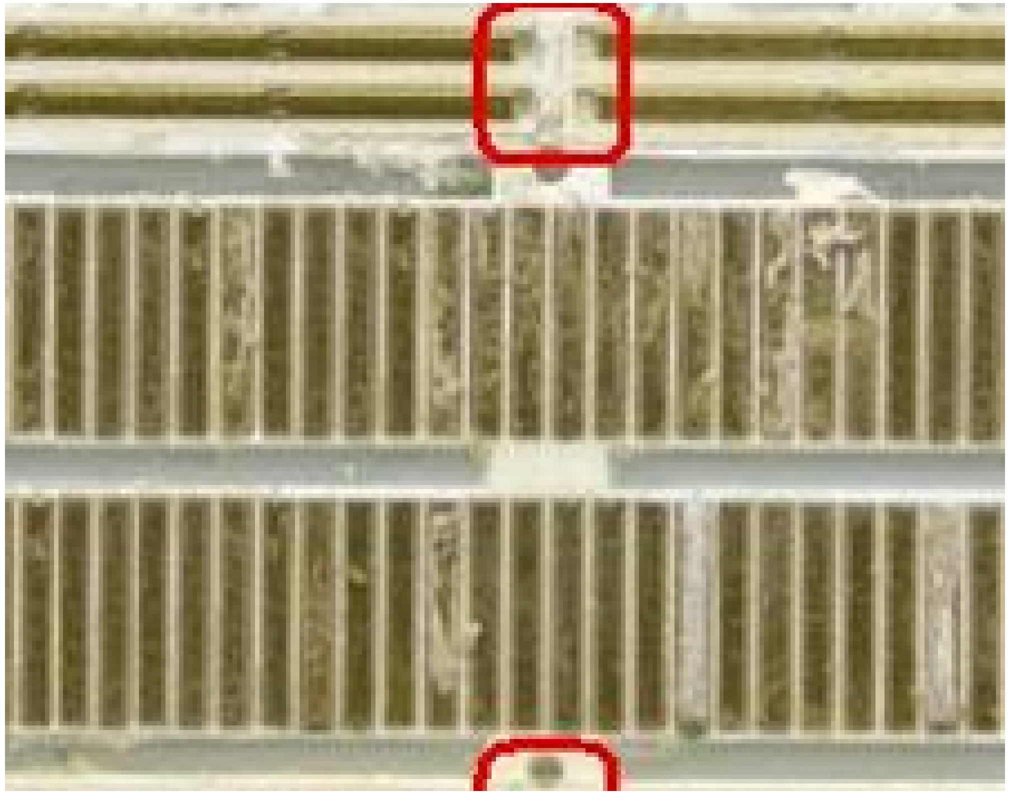
- For Electronics Prototyping

Outside	Inside
	

Breadboard Continued



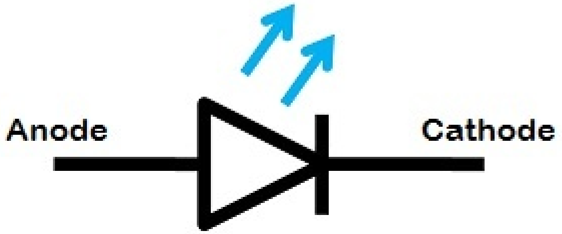
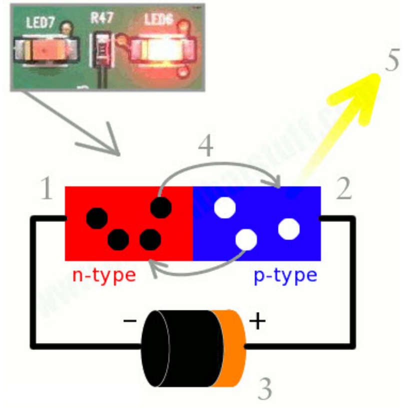
Breadboard Continued

Column1	Column2
	

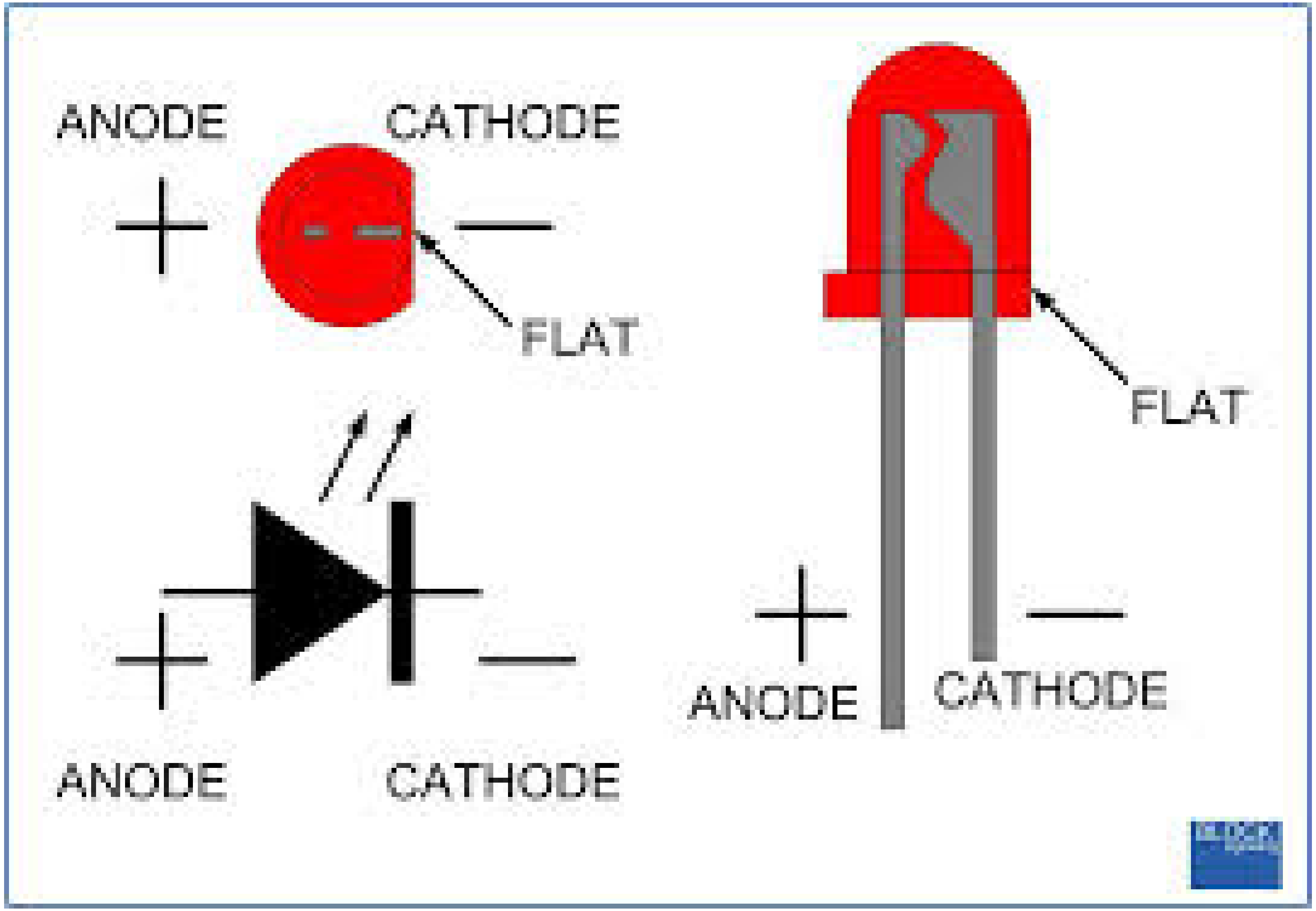
LED (Light Emitting Diode)

https://www.toppr.com/bytes/principles-of-led/

- Active Semiconductor Device

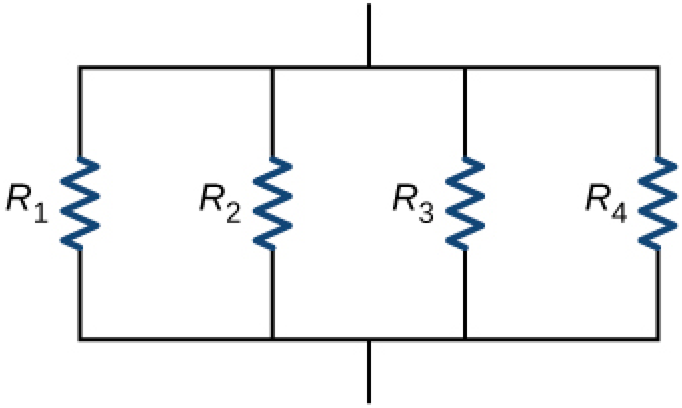
Column1	Column2	Column3
		

LED



Series and Parallel Resistance

(a) Resistors connected in series



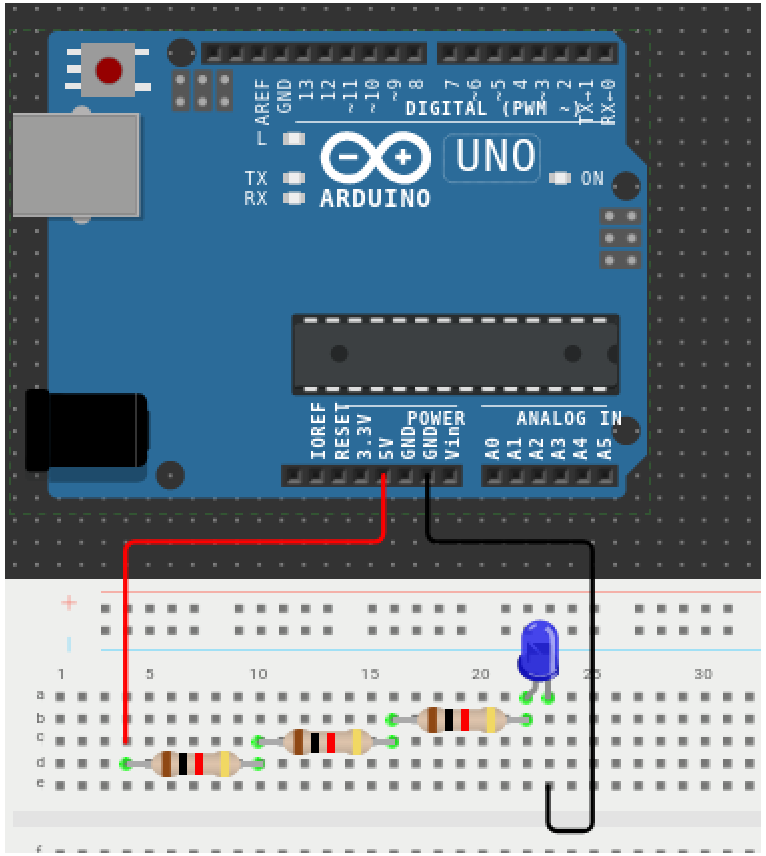
(b) Resistors connected in parallel

- Series Resistance $R_{eq} = R1 + R2 + R3 + R4$
- Parallel Resistance $\frac{1}{R_{eq}} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3} + \frac{1}{R4}$

Lets Build A Circuit

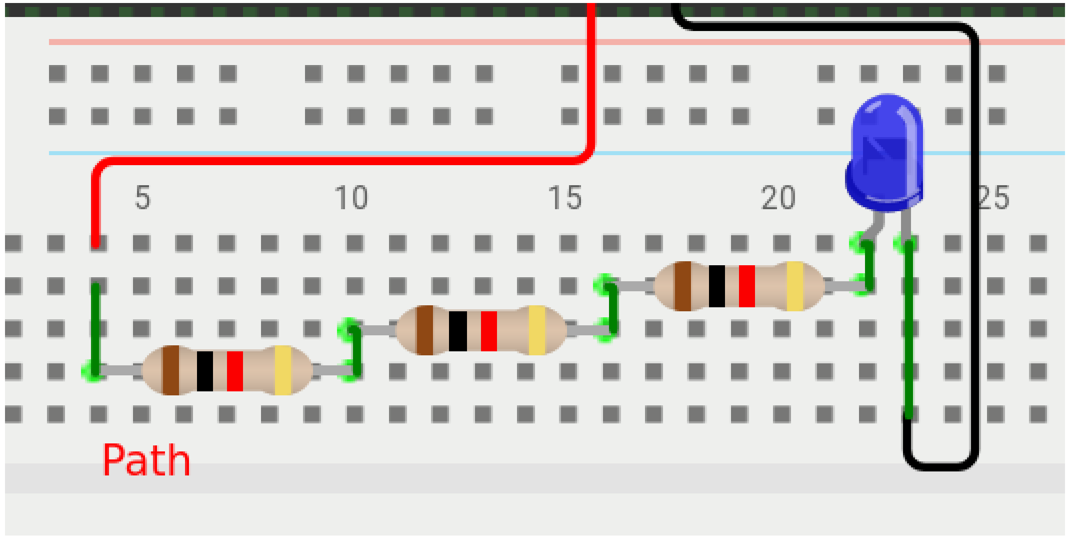
Series Resistance

https://wokwi.com/projects/376191541801099265

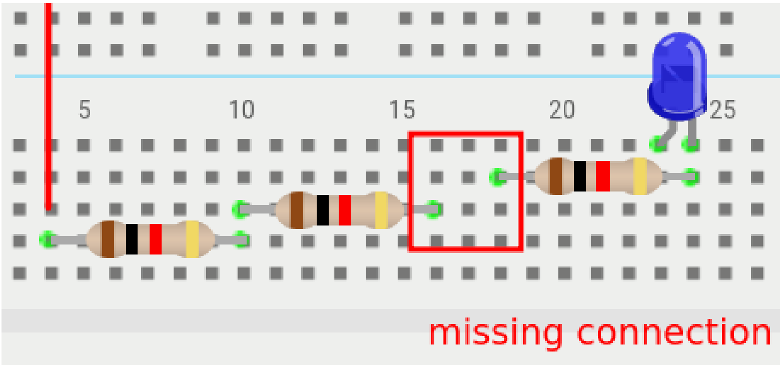
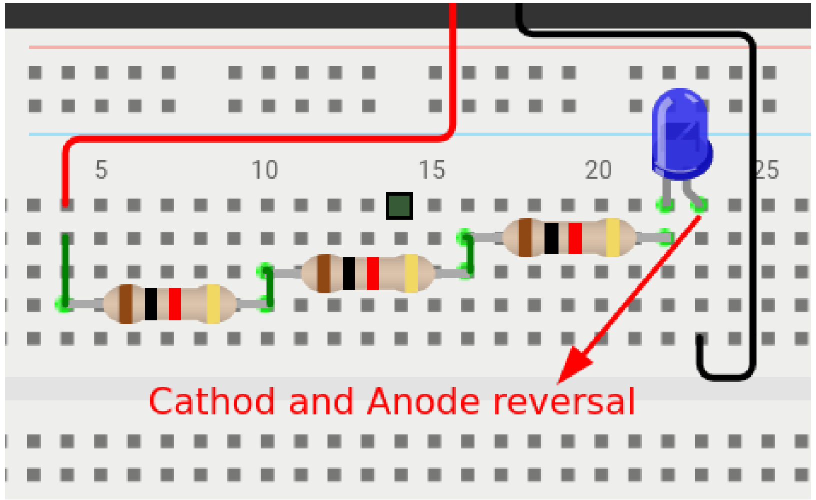
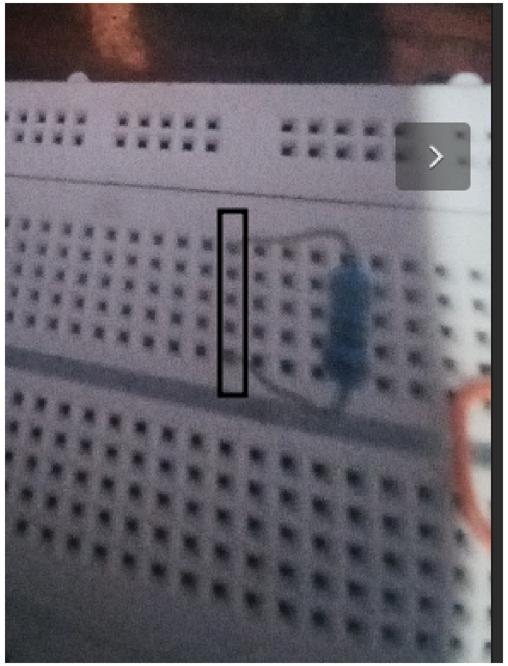


- Was there decrement in LED Glow?

Breadboard Path



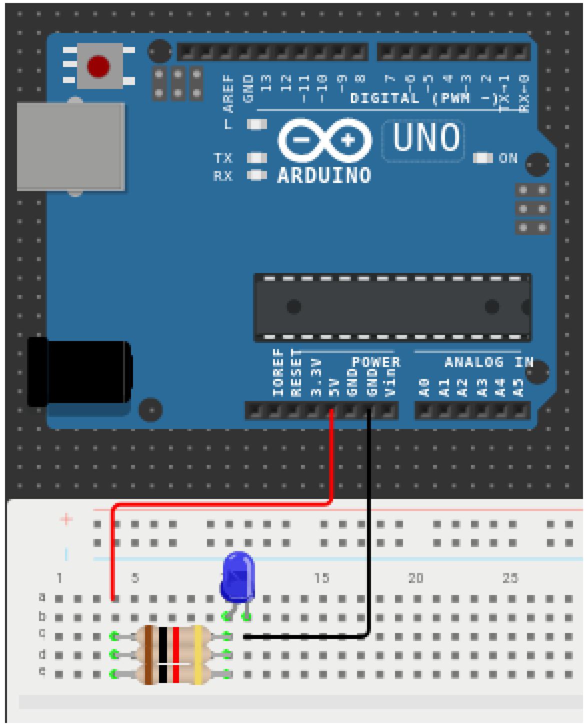
Common Mistakes

Lets Build A Circuit

Parallel Resitance

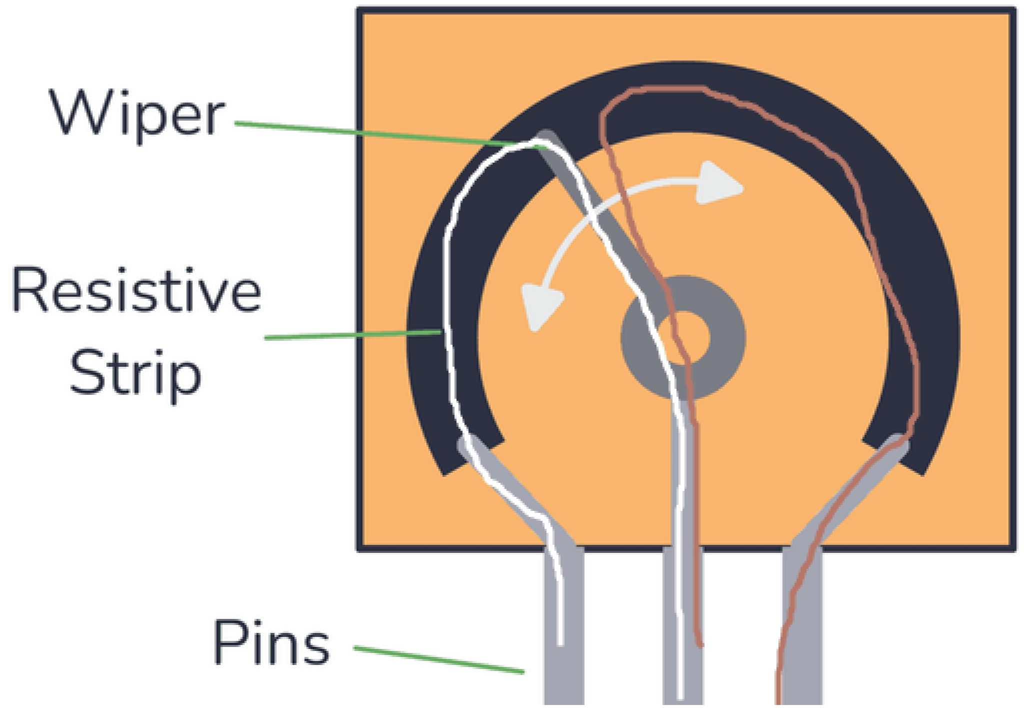
https://wokwi.com/projects/376191892088461313



- Was there increment in LED Glow?

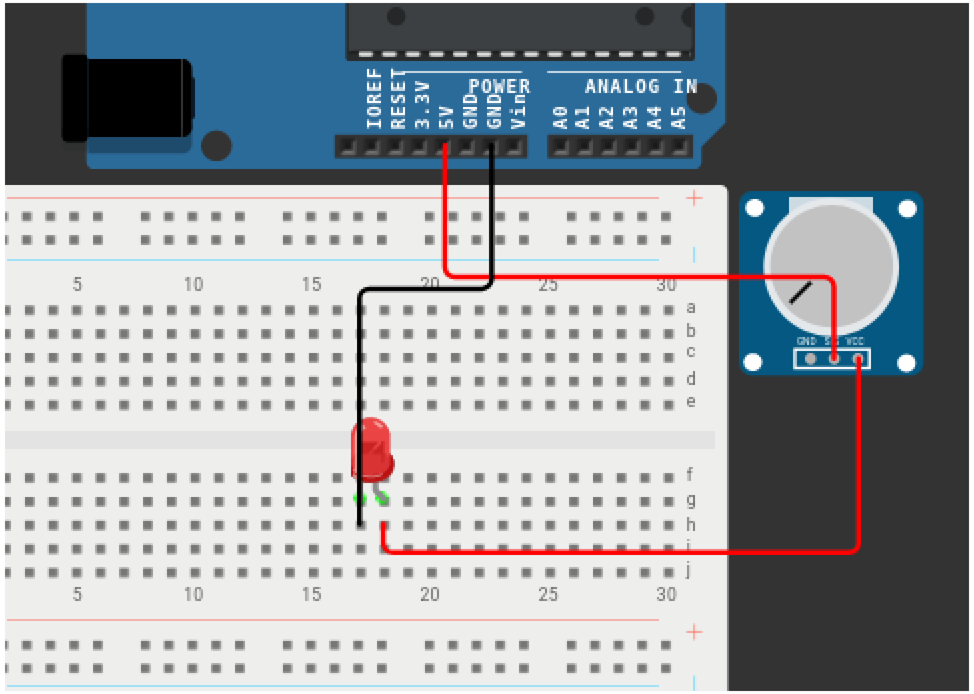
Potentiometer : Variabe Resitor

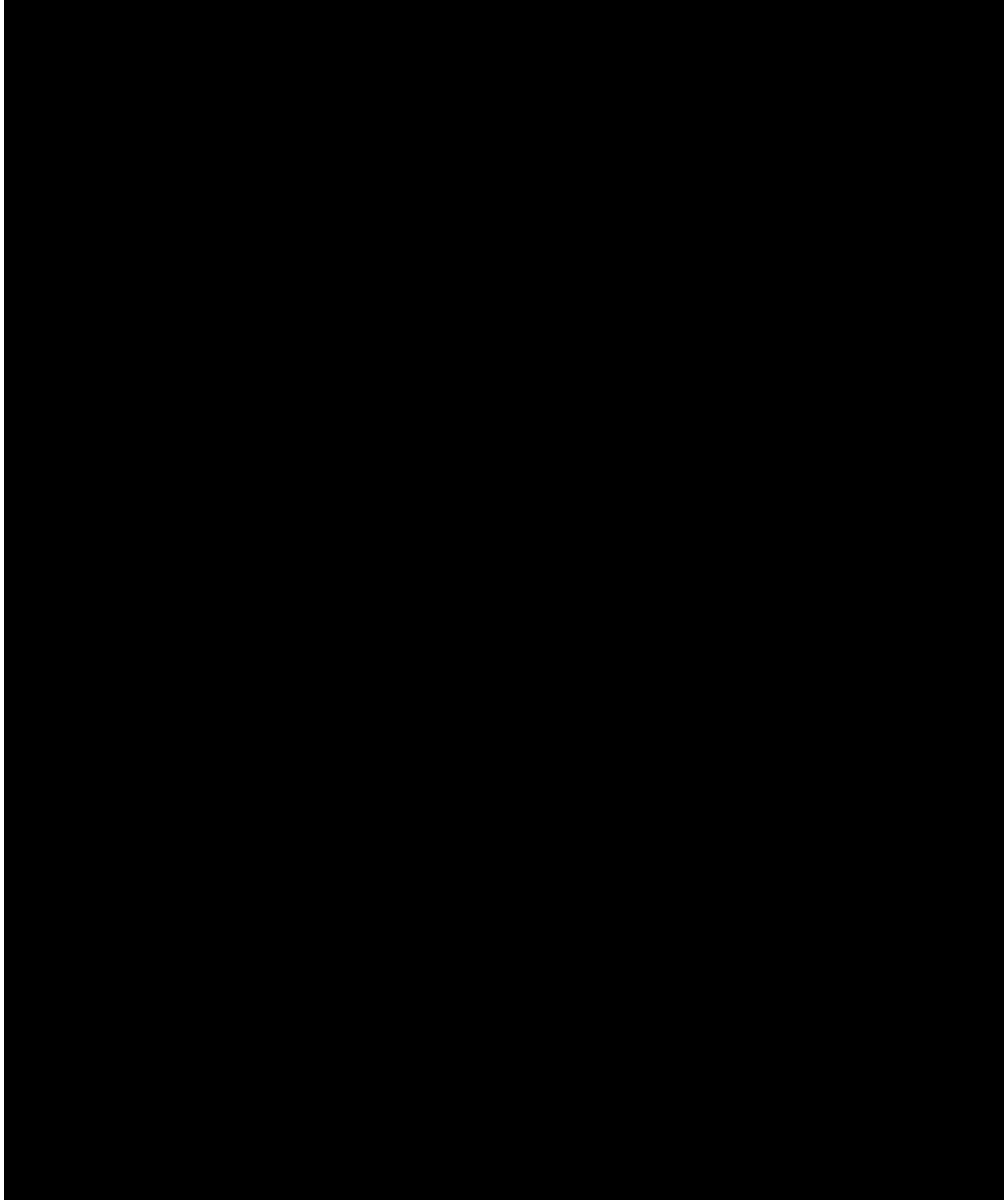
- $R = \rho \frac{L}{A}$



Build a circuit

- Potentiometer
  https://wokwi.com/projects/376209822976702465





Installing Arduino IDE

- For Linux : https://docs.arduino.cc/software/ide-v1/tutorials/Linux
- For Windows : https://docs.arduino.cc/software/ide-v1/tutorials/Windows
- For Mac : https://www.arduino.cc/en/Guide/macOS

Overview of writing code on Arduino IDE

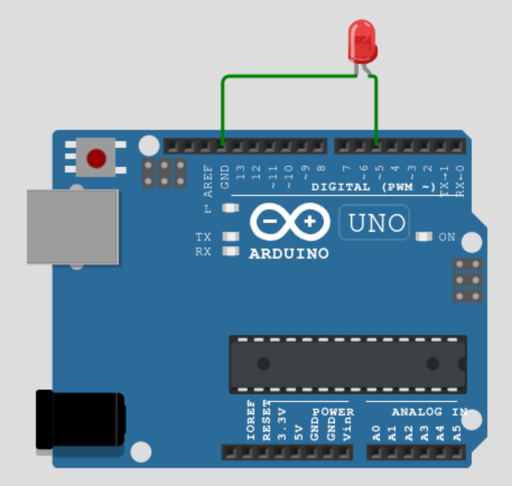
- Write code in Arduino IDE
- Connect Arduino to PC using USB cable
- Select the Port
- Compile the Program
- click button on IDE to load the program
- Done

Write Some Code on Arduino IDE

General Structure

```
void setup() {
        // setup code that runs once
}

void loop() {
        // main code that runs repeatedly
}
```

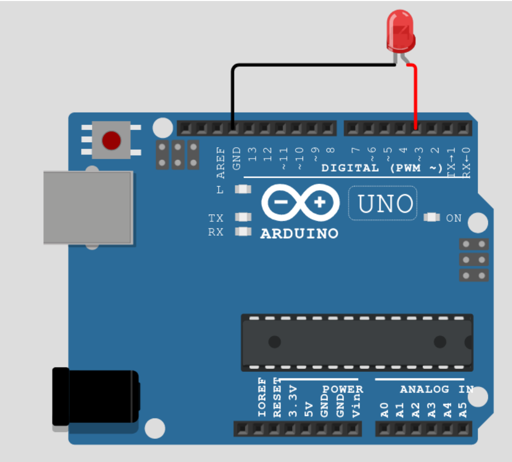
Example Code



Link to Simulator: https://wokwi.com/projects/338430633664578130

```
void setup() {
  pinMode(5,OUTPUT); //set pin no 5 on OUTPUT mode
}

void loop() {
  digitalWrite(5, HIGH); // send 5V
  delay(1000); // wait for 1 sec
  digitalWrite(5,LOW); // send 0 volts
  delay(1000); // wait for 1 sec
}
```

Example Code B



Link to Simulator: https://wokwi.com/projects/338430886227739218

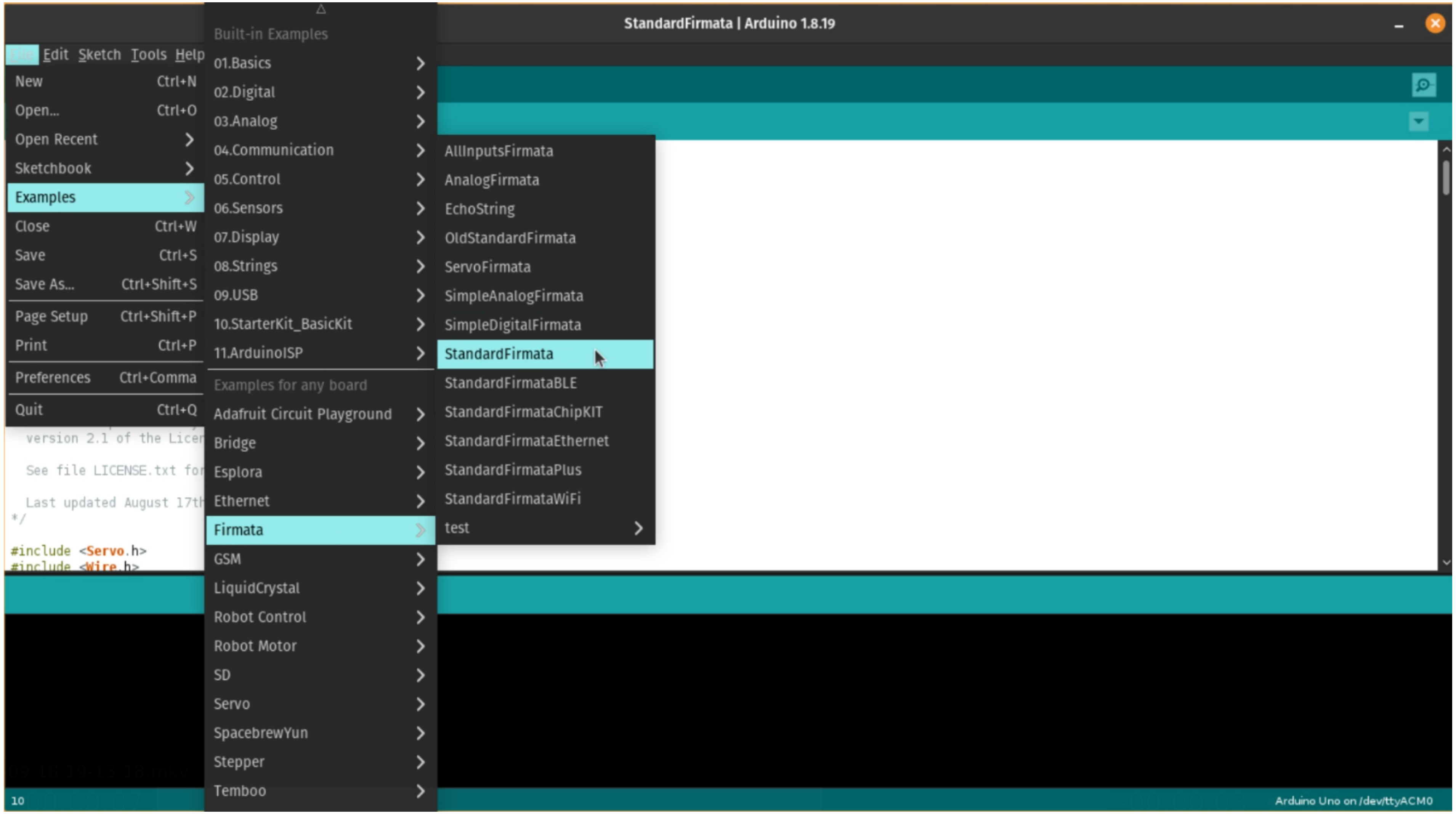
```
void setup() {
  pinMode(3, OUTPUT); // connected to PWM output
}

void loop() {
  analogWrite(3,255);  // to show how bright LED is at first
  delay(1000); // later fades away

  for(int i=0;i<255;i++){
    analogWrite(3,i); // analogWrite(pin, value) ; where value can range from 0 to 255.
    delay(100); // 0.1 sec delay
  }
}
```

Firmata

- Protocol for communicating with microcontrollers from software on a computer.
- All you need is to implement the protocol then any microcontroller can be supported.

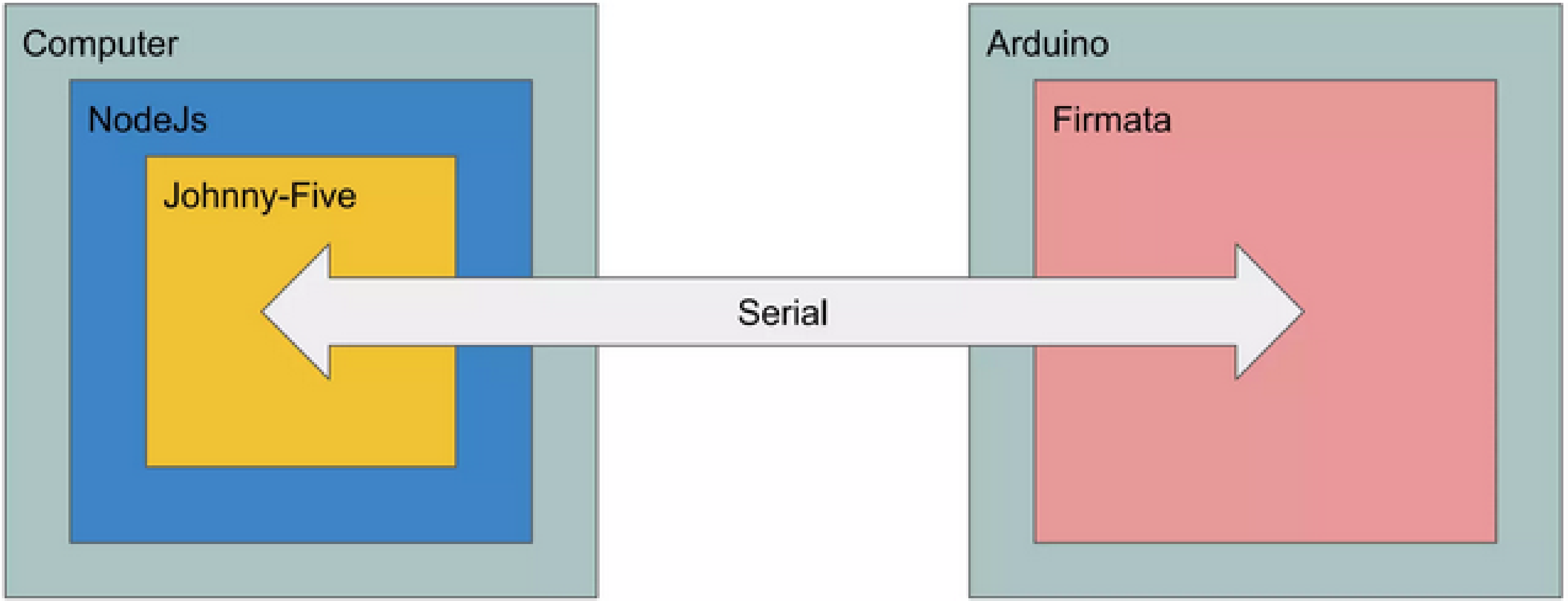
To Upload Firmata

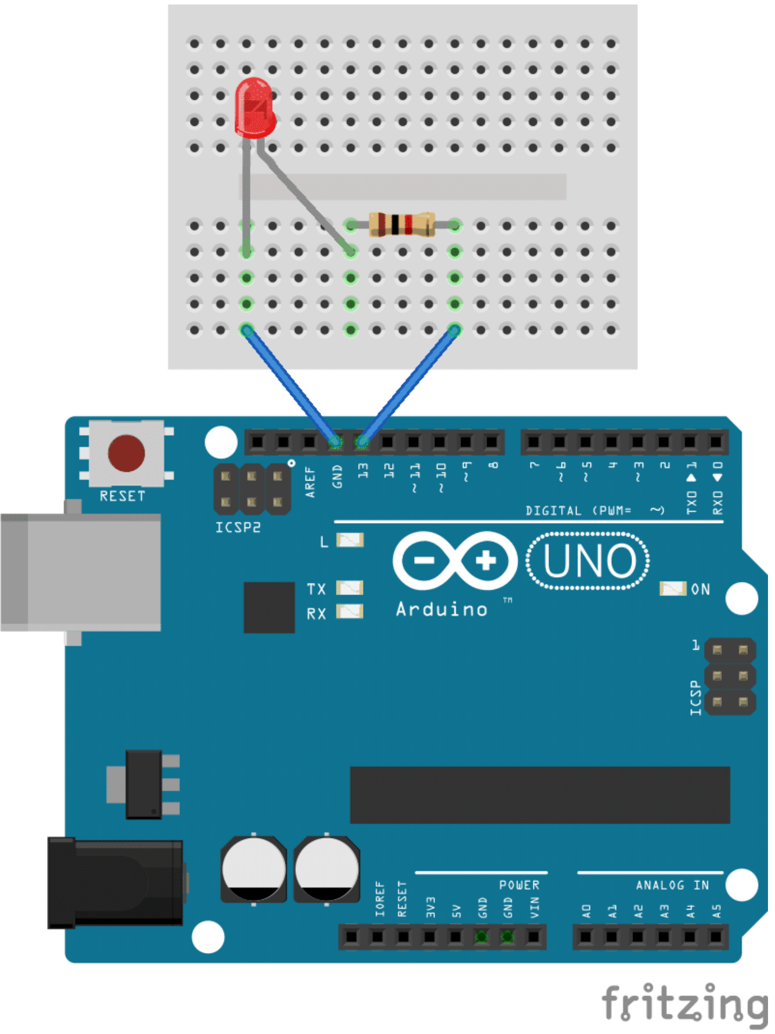
- connect the arduino to pc using USB cable
- open the arduino ide, select file>examples>firmata>StandardFirmata
- select port
- click upload


StandardFirmata | Arduino 1.8.19
Edit Sketch Tools Help
New Ctrl+N
Open... Ctrl+O
Open Recent
Sketchbook
Examples
Close Ctrl+W
Save Ctrl+S
Save As... Ctrl+Shift+S
Page Setup Ctrl+Shift+P
Print Ctrl+P
Preferences Ctrl+Comma
Quit Ctrl+Q
Built-in Examples
01.Basics
02.Digital
03.Analog
04.Communication
05.Control
06.Sensors
07.Display
08.Strings
09.USB
10.StarterKit_BasicKit
11.ArduinoISP
Examples for any board
Adafruit Circuit Playground
Bridge
Esplora
Ethernet
Firmata
GSM
LiquidCrystal
Robot Control
Robot Motor
SD
Servo
SpacebrewYun
Stepper
Temboo
AllInputsFirmata
AnalogFirmata
EchoString
OldStandardFirmata
ServoFirmata
SimpleAnalogFirmata
SimpleDigitalFirmata
StandardFirmata
StandardFirmataBLE
StandardFirmataChipKIT
StandardFirmataEthernet
StandardFirmataPlus
StandardFirmataWiFi
test
#include <Servo.h>
Arduino Uno on /dev/ttyACM0

Firmata

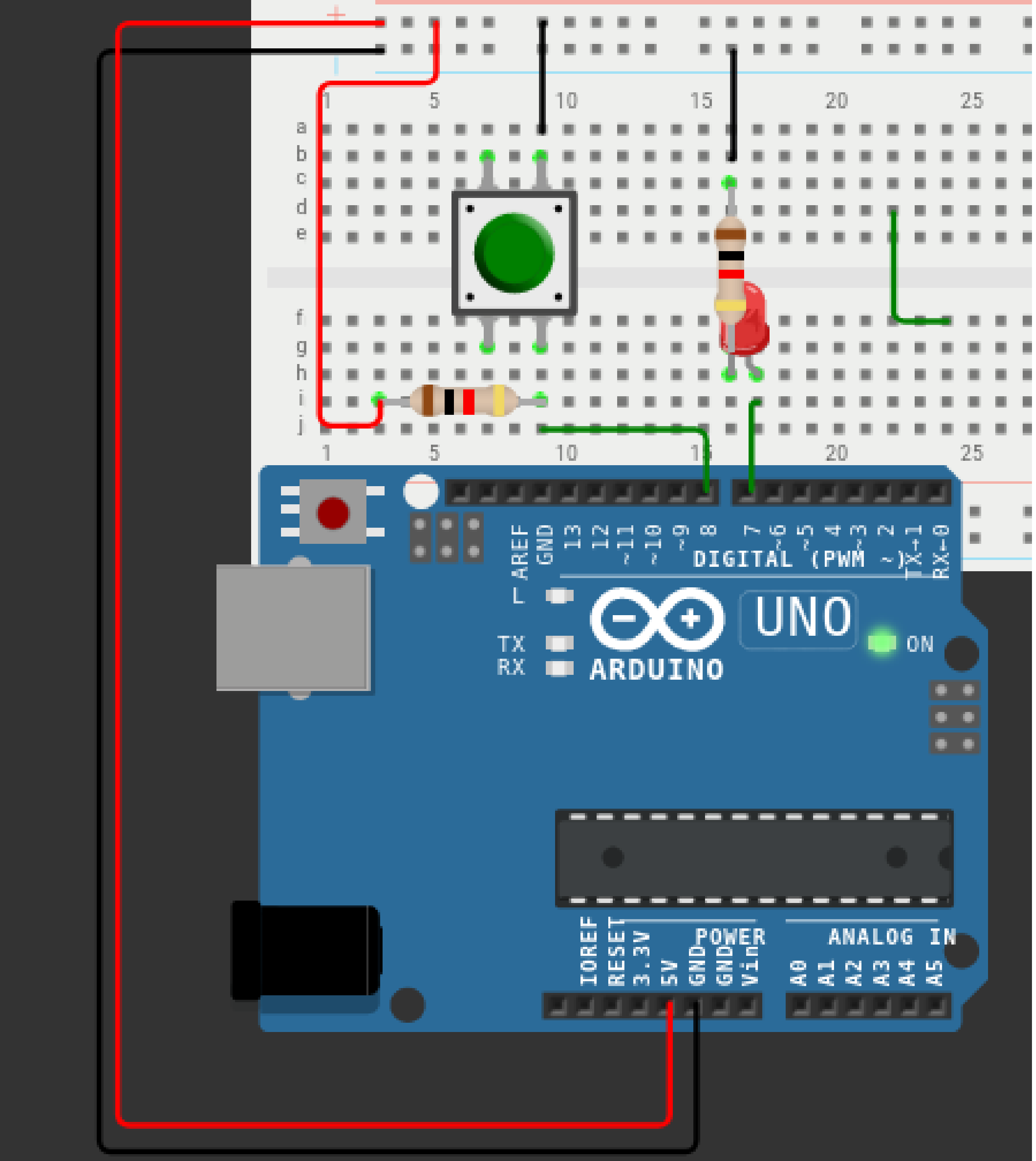
Basic Usage





Led Blink

```
var five = require("johnny-five");
var board = new five.Board();

board.on("ready", function() {
    var led = new five.Led(13);
    led.blink(500);
});
```

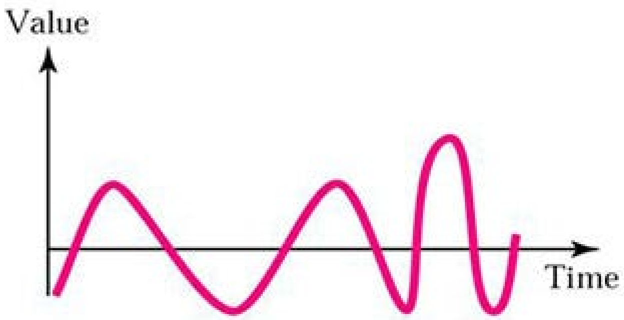


PushButton

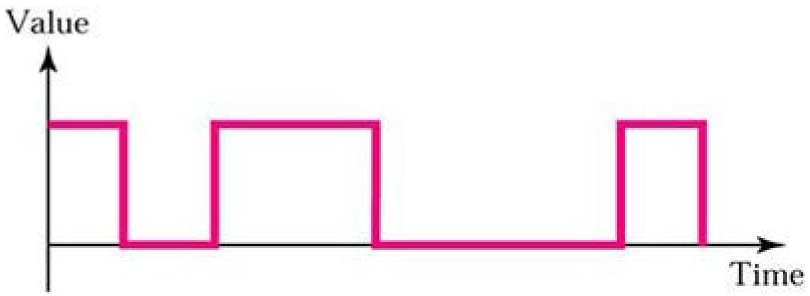
```
var five = require("johnny-five");
var board = new five.Board();

board.on("ready", function() {
   const led = new five.Led(7);
   const pushButton = new five.Button(
       {
           pin: 8,
           isPullup : true
       }
   );
   pushButton.on('down', () => {

       if (!led.isOn) {led.on();} else {led.off()}
   });

});
```

Analog Sensors

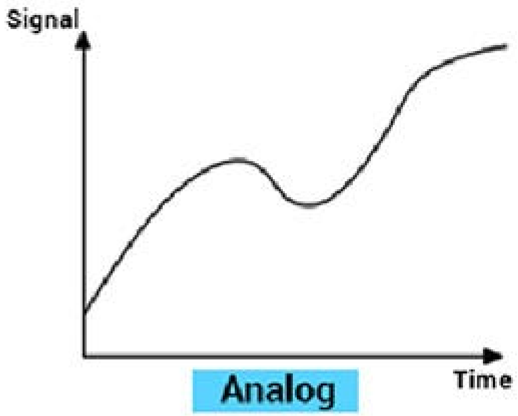
Analog Vs Digital

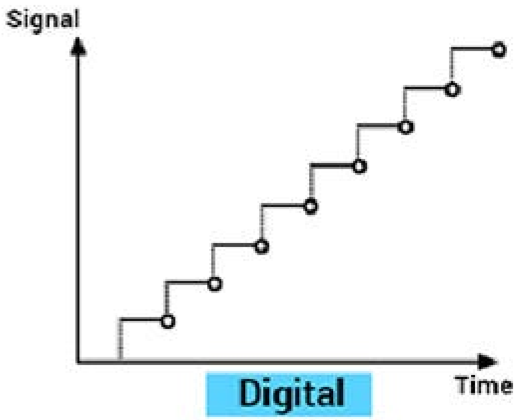


Analog signal



Digital signal

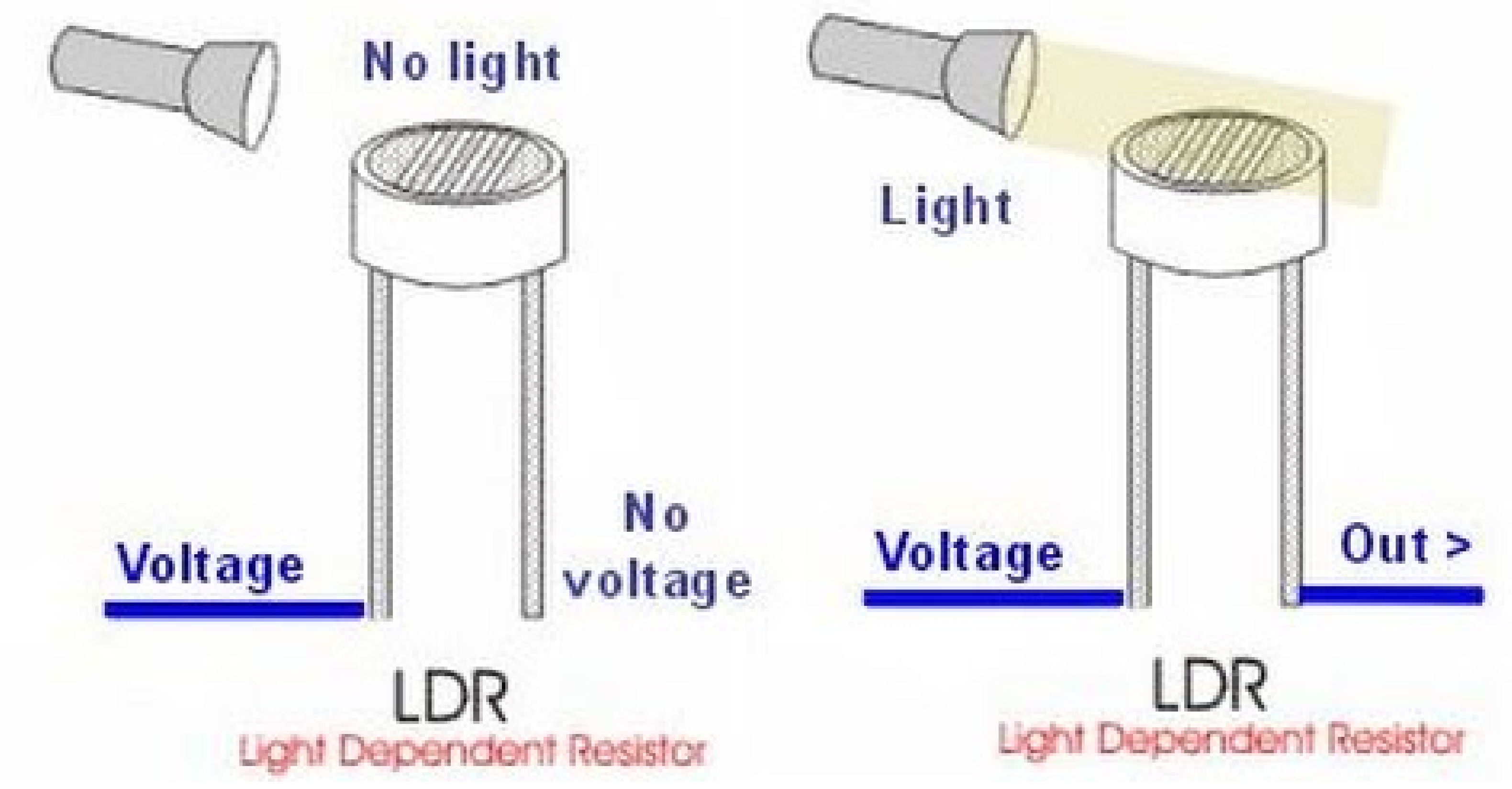


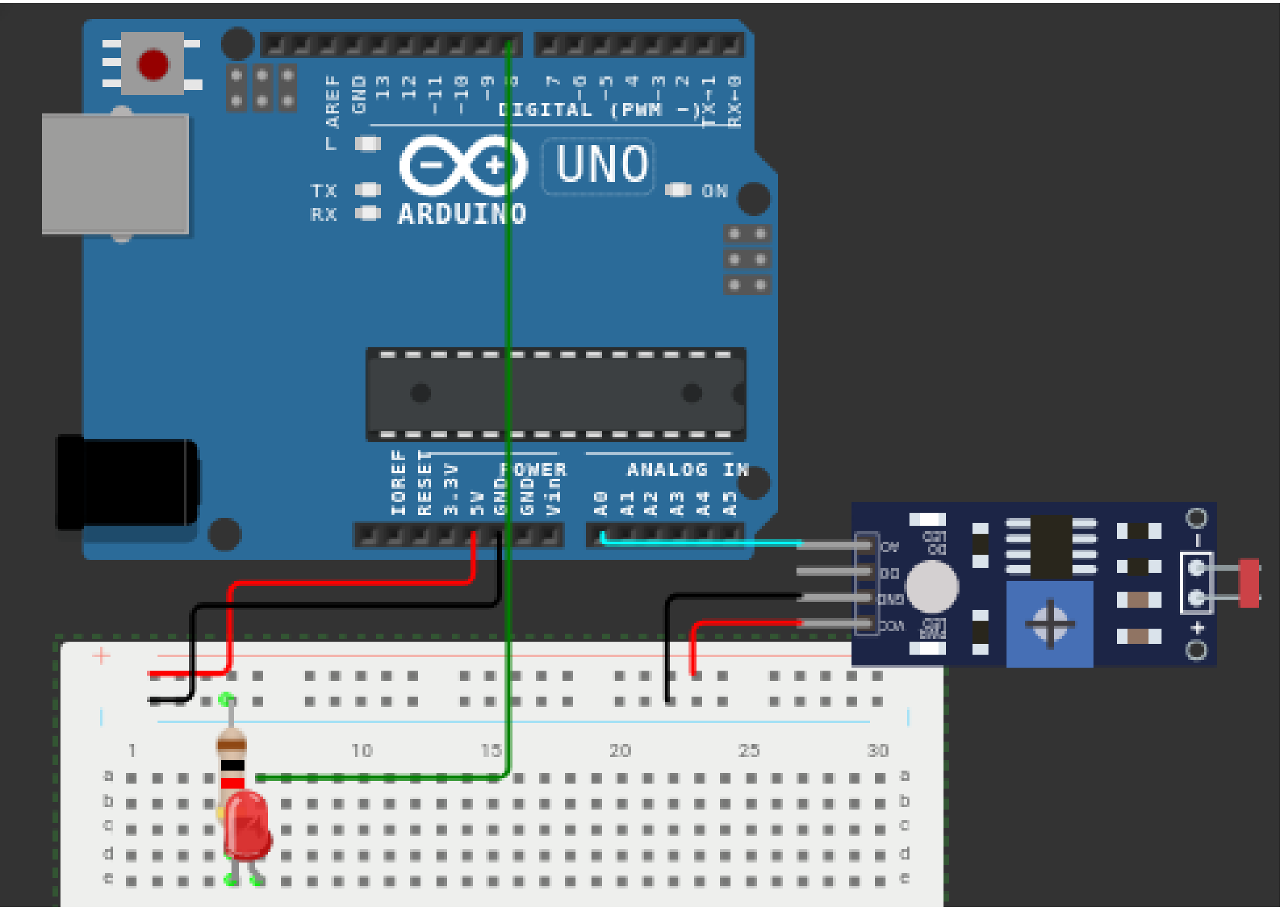


Analog Vs Digital

Aspect	Analog	Digital
Data Representation	Continuous waveform	Discrete binary values (0s and 1s)
Accuracy	High accuracy and precision	Excellent accuracy within bit resolution
Noise Resistance	Susceptible to noise and interference	More robust against noise
Signal Processing	Suitable for amplification, filtering, etc.	Versatile for complex algorithms
Cost and Complexity	Often simpler and potentially cheaper	Can be more complex and costly

LDR (Light Dependent Resistor)





https://johnny-five.io/api/sensor/

```
var five = require("johnny-five");

board = new five.Board();

board.on("ready", function() {

    let photoresistor = new five.Sensor({
        pin: "A0",
        freq: 500
    });

    let led = new five.Led(8);

    // board.repl.inject({
    //     pot: photoresistor
    // });

    photoresistor.on("data", function() {
        console.log(this.value);
        if (this.value >= 1000) {
            led.on();
        }else {
            led.off();
        }
    });
});
```

Equivalent C code

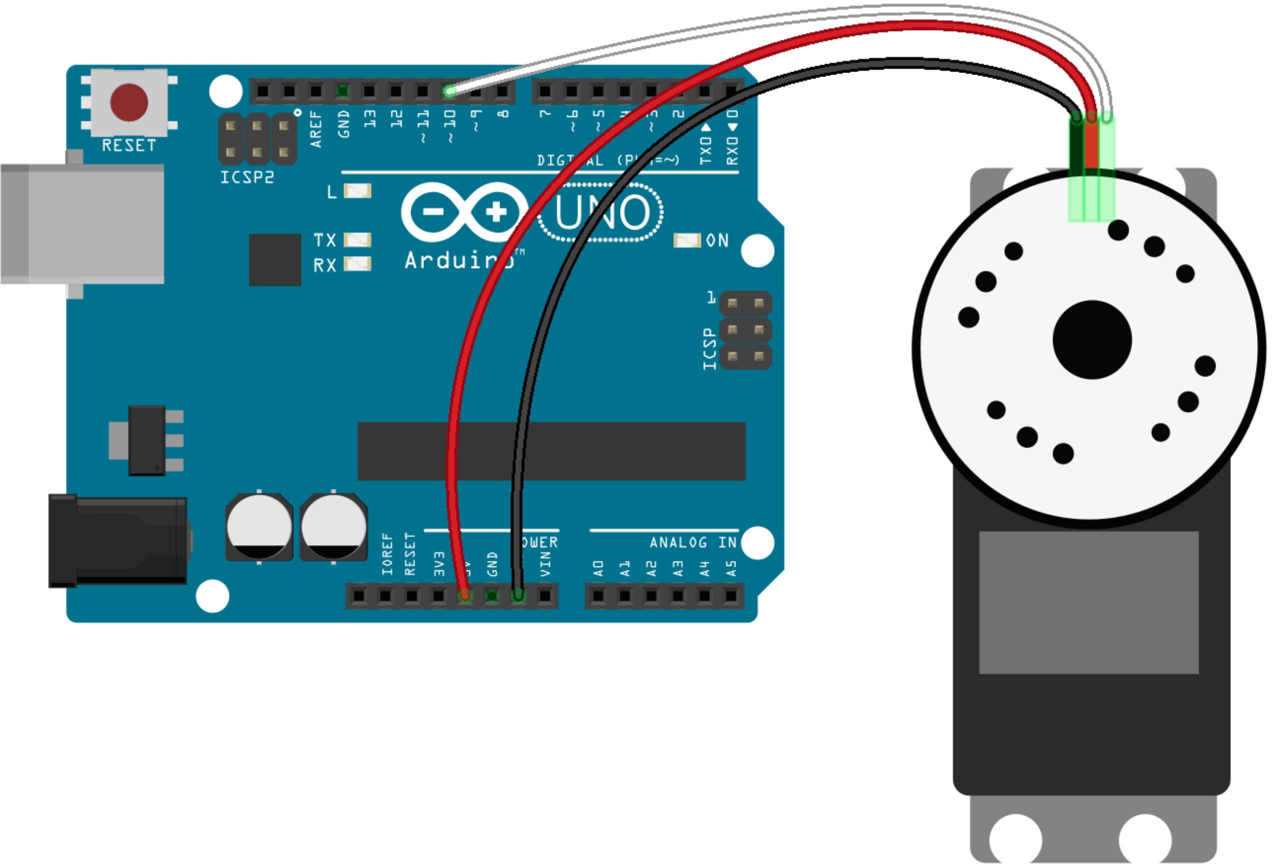
https://wokwi.com/projects/376361780745305089

```
#define LDR A0
#define LED 8

void setup() {
pinMode(8, OUTPUT);
pinMode(A0, INPUT);

}

void loop() {
  if (analogRead(A0) >= 1000) {
    digitalWrite(LED, HIGH);
  }else {
    digitalWrite(LED, LOW);
  }
}
```

Servo Control

Motors

Motor Type	Description	Image
Stepper	Converts electrical pulses into precise mechanical steps. Used in 3D printers and CNC machines.	
Servo	Provides accurate angular control, common in robotics.	
DC	Operates on direct current, simple, and used in various applications.	



Servo Motor

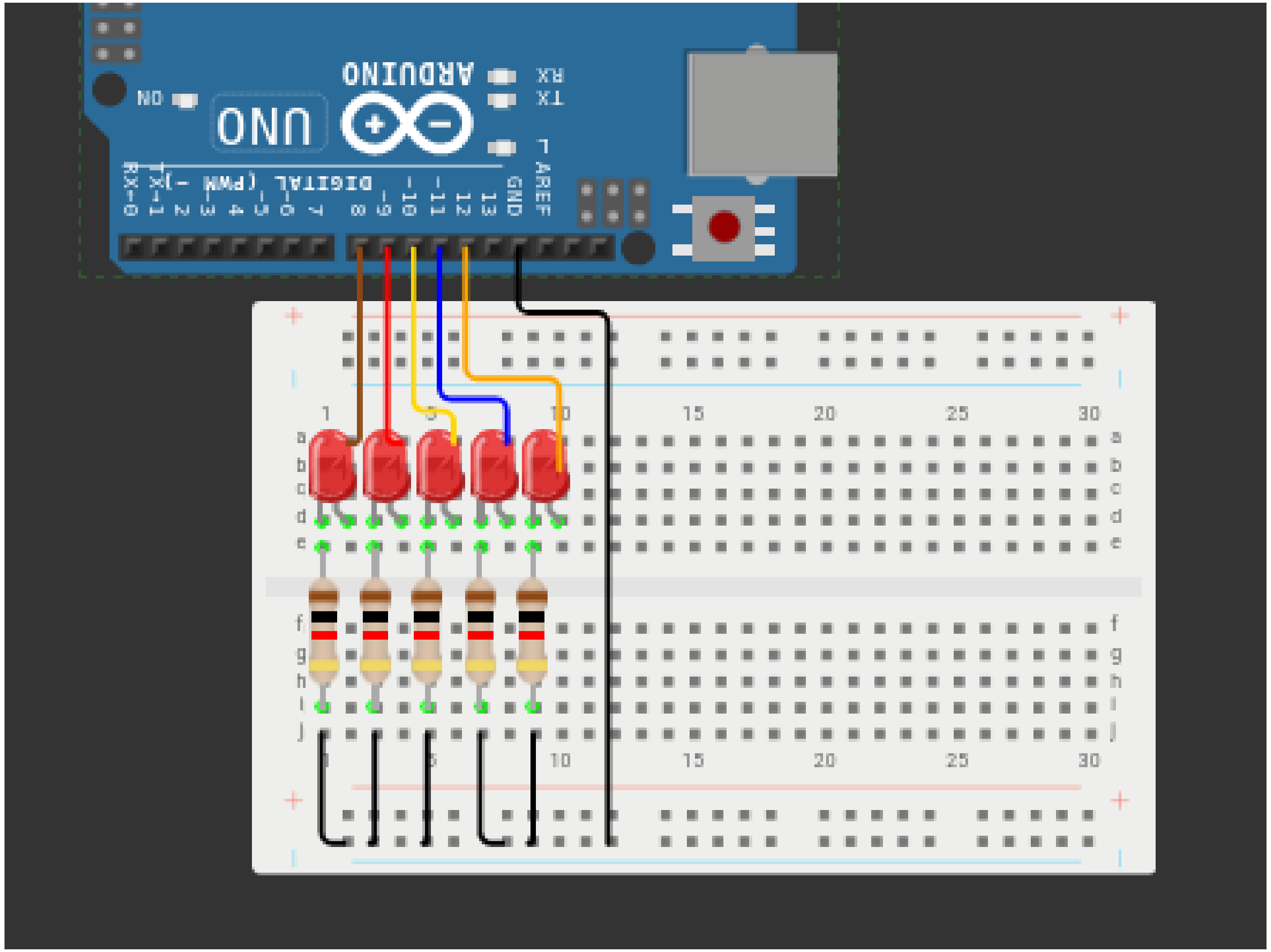
```
const five = require("johnny-five");
const board = new five.Board();

board.on("ready", () => {
    const servo = new five.Servo(10);

    board.repl.inject({
        servo
      });
    servo.sweep();
})
```

Build a project

LED Binary Counter

Circuit

https://wokwi.com/projects/376380254565660673



```
const five = require("johnny-five");
const board = new five.Board();


board.on("ready", () => {
    let leds = [8,9,10,11,12].map((pinNo) => {
       return new five.Led(pinNo);
    });

    for (let i = 0 ; i <= 31; i++) {
        setTimeout(() => {ledState(i.toString(2).padStart(5,'0') ,leds)},i * 1000);
    }


}) ;

function ledState(ledString,leds) {
    Array.from("00000").map((val,index) => setLedState(leds[index], parseInt(val)));
    Array.from(ledString).map((val,index) => setLedState(leds[index], parseInt(val)));
}

function setLedState(led, state) {
    state == 1 ? led.on() : led.off();
}
```

Thankyou